

Thermodynamics

Rankine cycle:

The Rankine cycle is a thermodynamic cycle that describes the operation of steam power plants. It is the most widely used cycle for electricity generation.

Key Components:

1- Boiler / Steam Generator 2- Turbine
3- Condenser 4- Pump

Process of Rankine Cycle:

State 1-2: Reversible adiabatic pumping process in pump

State 2-3: Constant Pressure transfer of heat in boiler

State 3-4: Reversible adiabatic expansion in turbine

State 4-1: Constant pressure transfer of heat in condenser

Efficiency:

$$\eta = \frac{W_{net}}{Q_{in}}$$

Example #9.1

To determine the efficiency of cycle, we will go through the problem step by step

Step 1: Work done by the Pump

$$W_p = v(P_2 - P_1) \quad \text{--- (1)}$$

$$P_1 = 10 \text{ kPa (given)}$$

$$\text{At } P_1; v = 0.001010 \text{ m}^3/\text{kg} \quad (v_f \text{ because of liquid state})$$

$$P_2 = 2 \text{ MPa}$$

$$\text{put values } W_p = (0.00101) (2 \times 10^6 - 10 \times 10^3) = \boxed{2.009 \text{ kJ/kg}}$$

Specific enthalpy

$$W_p = h_2 - h_1$$

$$h_2 = h_1 + W_p$$

$$\text{At } P = 10 \text{ kPa}, h_1 = 191.8 \text{ kJ/kg}$$

$$\text{So, } h_2 = 191.8 \text{ kJ/kg} + 2 \text{ kJ/kg}$$

$$\boxed{h_2 = 193.8 \text{ kJ/kg}}$$

Step 2: Heat added in Boiler (q_H)

$$q_H = h_3 - h_2$$

$$\text{At } P = 2 \text{ MPa}; h_3 = 2799.51 \text{ kJ/kg}$$

(vg because vapor state)

$$q_H = 2799.51 - 193.8$$

$$\boxed{q_H = 2605.7 \text{ kJ/kg}}$$

Step 3: Work output of turbine (W_t)

$$* W_t = h_3 - h_4 \quad \text{--- (2)}$$

h_4 can be found by entropy conditions

$$\text{Let } S_3 = S_4$$

$$\text{At } P = 2 \text{ MPa}; S_3 = 6.3408$$

(sat) (Table B.1.3)

$$S_4 = 6.3408$$

$$\text{Now at } P_1 = 10 \text{ kPa}$$

$$S_f = 0.6492; S_{fg} = 7.5010$$

So for quality

$$S_4 = S_f + x_4 \cdot S_{fg}$$

$$x = \frac{6.3408 - 0.6492}{7.5010} = 0.752$$

As S_4 is back in liquid state that's why we have taken S_f, S_{fg} at $P_1 = 10 \text{ kPa}$ and not $P_2 = 2 \text{ MPa}$

$$\text{Now, } h_4 = h_f + x \cdot h_{fg}$$

$$\text{At } P = 10 \text{ kPa}$$

$$h_f = 191.8 \text{ kJ/kg}, h_{fg} = 2392.82$$

$$\text{So, } h_4 = 1991.21 \text{ kJ/kg}$$

So, equation (2) will be

$$W_t = 2799.51 - 1991.21$$

$$\boxed{W_t = 808.29 \text{ kJ/kg}}$$

* in this part values doesn't match with book but they're right according to the table

Step 4: Heat rejected in condenser (q_L)

$$\text{Now, } q_L = h_4 - h_1$$

$$q_L = 1991.21 - 191.8$$

$$\boxed{q_L = 1799.41 \text{ kJ/kg}}$$

Thermal efficiency:

$$\eta_{th} = \frac{W_{net}}{q_H}$$

$$W_{net} = W_t - W_p$$

$$= 808.29 - 2.0 =$$

$$\boxed{W_{net} = 806.29 \text{ kJ/kg}}$$

$$q_H = 2605.7 \text{ kJ/kg}$$

$$\eta_{th} = \frac{806.29}{2605.7} \times 100$$

$$\boxed{\eta_{th} = 30.94\%}$$

9.3 Effect of Pressure and Temperature on Rankine Cycle

- Effect of increasing boiler pressure

- ↳ higher enthalpy (h_3)
- ↳ more heat input
- ↳ Thermal efficiency increases $\eta = \frac{T_{high} - T_{low}}{T_{high}}$
- ↳ Work output increases
- ↳ Heat rejection decreases

- Effect of increasing Boiler Temperature

- ↳ Thermal efficiency increases $q_H = h_3 - h_2$; $W_t = h_3 - h_4$
- ↳ Work output increases
- ↳ Lower moisture content

- Effect of Reducing condenser Pressure

- ↳ Efficiency increases
- ↳ Work output increases
- ↳ Large cooling systems

- Effect of Reheating

- ↳ Efficiency increases
- ↳ Work output increases
- ↳ Complex cycle design, additional equipment

Example 9.2

$$w_p = v(P_2 - P_1)$$
$$= (0.00101)(4000 - 10)$$

$$w_p = 4 \text{ kJ/kg}$$

Now,

$$h_1 = 191.8 \text{ kJ/kg}$$

$$h_2 = 191.8 + 4 = 195.8 \text{ kJ/kg}$$

Step 2:

$$w_t = h_3 - h_4 ; s_4 = s_3$$

$$h_3 = 3123.6 \text{ kJ/kg} \text{ by table}$$

$$s_3 = 6.7690 \text{ kJ/kg} \text{ by table}$$

$$6.7690 = 0.60193 + x_4(7.5009)$$

$$x_4 = 0.8159$$

$$h_4 = 191.8 + (0.8159)(2392.8)$$

$$h_4 = 2144.1 \text{ kJ/kg}$$

Now

$$w_t = 3123.6 - 2144.1$$

$$w_t = 1069.5 \text{ kJ/kg}$$

$$\text{Now } w_{\text{net}} = w_t - w_p$$
$$= 1069.5 - 4$$

$$w_{\text{net}} = 1065.5 \text{ kJ/kg}$$

Final Step:

$$q_H = h_3 - h_2$$
$$= 3123.6 - 195.8$$

$$q_H = 3017.8 \text{ kJ/kg}$$

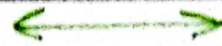
$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_H} = \frac{1065.5}{3017.8} = 35.3\%$$

$$q_L = h_4 - h_1$$
$$= 2144.1 - 191.8$$

$$q_L = 1952.3 \text{ kJ/kg}$$

$$w_{\text{net}} = q_H - q_L$$
$$= 3017.8 - 1952.3$$

$$w_{\text{net}} = 1065.5 \text{ kJ/kg}$$



End Problem 9.13

data:

$$\text{Boiler Pressure} = 3 \text{ MPa}$$

$$\text{Condenser Pressure} = 10 \text{ kPa}$$

State 1: Pump Inlet

$$\text{At } P = 10 \text{ kPa}$$

$$h_1 = 191.8 \text{ kJ/kg} ; v_1 = 0.00101 \text{ m}^3/\text{kg}$$

both for liquid state

$$\text{Now, } w_p = v(P_2 - P_1)$$

$$w_p = (0.00101)(3000 - 10)$$

$$w_p = 3.02 \text{ kJ/kg}$$

$$\text{Now, } h_2 = h_1 + w_p$$

$$h_2 = 191.8 + 3.02$$

$$h_2 = 194.82 \text{ kJ/kg}$$

Step 2: Boiler Inlet

$$\text{At } P = 3 \text{ MPa (saturated vapor)}$$

$$h_3 = 2804.14 \text{ kJ/kg}$$

$$s_3 = 6.1869 \text{ kJ/kg}$$

Step 3: Turbine Inlet

$$s_4 = s_3 = 6.1869 \text{ kJ/kg}$$

Now at $P = 10 \text{ kPa}$

$$s_g = 0.6492 \text{ kJ/kg}$$

$$s_{fg} = 7.5010 \text{ kJ/kg}$$

Now for quality

$$x_u = \frac{0.6492 + 6.1869}{7.5010}$$

$$x_u = 0.738$$

So,

$$h_u = (h_f) + x h_{fg}$$

$$h_u = (191.8) + (0.738)(2392.8)$$

$$h_u = 1957.68 \text{ kJ/kg}$$

Now first we will find q_H

$$q_H = h_3 - h_2$$
$$= 2804.14 - 194.82$$

$$q_H = 2609.32 \text{ kJ/kg}$$

Now we will find

$$w_t = h_3 - h_u$$
$$= 2804.14 - 1957.68$$

$$w_t = 846.46 \text{ kJ/kg}$$

Step 4: Condenser

$$q_L = h_u - h_1$$
$$= 1957.68 - 191.8$$

$$q_L = 1765.88$$

Efficiency:

$$\eta_{th} = \frac{W_{net}}{q_H}$$

$$= \frac{w_t - w_p}{q_H} \times 100$$

$$= \frac{846.46 - 3.02}{2609.32} \times 100$$

$$\eta_{th} = 32.32\%$$

Rankine Cycle Questions

Step 1: Pump

① At P (sat liquid) find h and v
then find $w_p = v(P_2 - P_1)$ and

$$h_2 = h_1 + w_p$$

Step 2: Boiler

At P (sat vapor) find h_3 and s_3
and then $q_H = h_3 - h_2$

Step 3: Turbine

Make $s_3 = s_4$, $P = (\text{sat liq})$
find s_f and s_{fg} , then x

Use x to find h_4 at
(sat vapor) h_f and h_{fg}
then find $w_t = h_3 - h_4$

Step 4: Condenser

$$q_L = h_u - h_1$$

Step 5: Efficiency

$$\eta = \frac{W_{net}}{q_H} = \frac{w_t - w_p}{q_H} \times 100$$